VERIFICATION CERTIFICATE FOR THE LEAN 4 DEVELOPMENT
TORUS-VELOCITY-TETHER

BENJAMIN FROHMAN

ABSTRACT. This document certifies that the Lean 4 formalization in the repository `torus-velocity-tether` builds with its pinned toolchain, contains no `sorry` and no user-declared axioms, and that every theorem depends only on Lean's standard axioms.

1. VERIFICATION CERTIFICATE

The formal development was machine-checked as follows.

- Repository:
  `https://github.com/BenFrohman/torus-velocity-tether`
- Verified commit:
  `0a916ec06e17780af0788a7b3bf8e034ff7bf107`
- Lean toolchain: `leanprover/lean4:v4.30.0-rc1`
- Mathlib: tag `v4.30.0-rc1`, commit (from `lake-manifest.json`)
  `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  `https://github.com/BenFrohman/torus-velocity-tether/actions/runs/41571865074`
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  `https://github.com/BenFrohman/torus-velocity-tether/archive/0a916ec06e17780af0788a7b3bf8e034ff7bf107.tar.gz`
  `022a2f44b68c4bd17a14aecb74dce6dada1e4675a90184ef940458a26af10f06`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 63 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`TorusVelocityTether.IsPhase.conj`	`[propext, Classical.choice, Quot.sound]`
`TorusVelocityTether.Pi_u_act`	`[propext, Classical.choice, Quot.sound]`
`TorusVelocityTether.Pi_u_self`	`[propext, Classical.choice, Quot.sound]`
`TorusVelocityTether.Pi_u_smul`	`[propext, Classical.choice, Quot.sound]`
`TorusVelocityTether.Pi_u_smul_self`	`[propext, Classical.choice, Quot.sound]`
`TorusVelocityTether.TorusState.prop`	`[propext, Classical.choice, Quot.sound]`

Date: November 2, 2026.

Lean declaration	#print axioms
TorusVelocityTether.actFun_conj_actFun	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.actState.congr_simp	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.actState_	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.act_act_conj	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.act_add	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.act_add_smul_conj	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.act_conj_act	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.act_smul	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.act_sub	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.c1_c3_not_unique_T	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.comparison_ode_ceiling	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.conj_mul_self_of_norm_one	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.cubic_field_neg	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.eq_zero_of_fieldPairing_self	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.fd_act	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.fieldCurl_act	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.fieldCurl_smul	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.fieldPairing_act	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.fieldPairing_self	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.fieldPairing_smul_left	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.fieldPairing_smul_right	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.finEnergy_act	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.isL2_act	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.isRep_act	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.kineticEnergyCurlIdentity	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.modeDensity_strict_neg	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.modeE_act	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.modePairing_act_conj_right	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.modePairing_add_add	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
TorusVelocityTether.modePairing_comm	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.modePairing_phase	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.modePairing_self	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.modePairing_self_nonneg	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.modePairing_self_pos	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.modePairing_smul_left	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.modePairing_smul_right	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.scaleState_	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.step2_degree_T	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.step5_higher_order_vanishes	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.step5_tetherKernelT	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.tetherKernelT_C1	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.tetherKernelT_C2	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.tetherKernelT_C3	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.tetherKernelT_comm	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.tetherKernelT_diag_nonpos	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.tetherKernelT_energy_conservation	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.tetherKernelT_feedbackAtMost	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.tetherKernelT_polarizes	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.tetherKernelT_polarizes_at	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.tetherKernelT_quadratic_form	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.torus_velocity_tether_product_theorem	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.uniqueness_T	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.velSlot_act	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.velSlot_energy	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.velSlot_scale	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.wt_nonneg	[propext, Classical.choice, Quot.sound]
TorusVelocityTether.young_absorption	[propext, Classical.choice, Quot.sound]

Email address: frohmanbenjamin@gmail.com